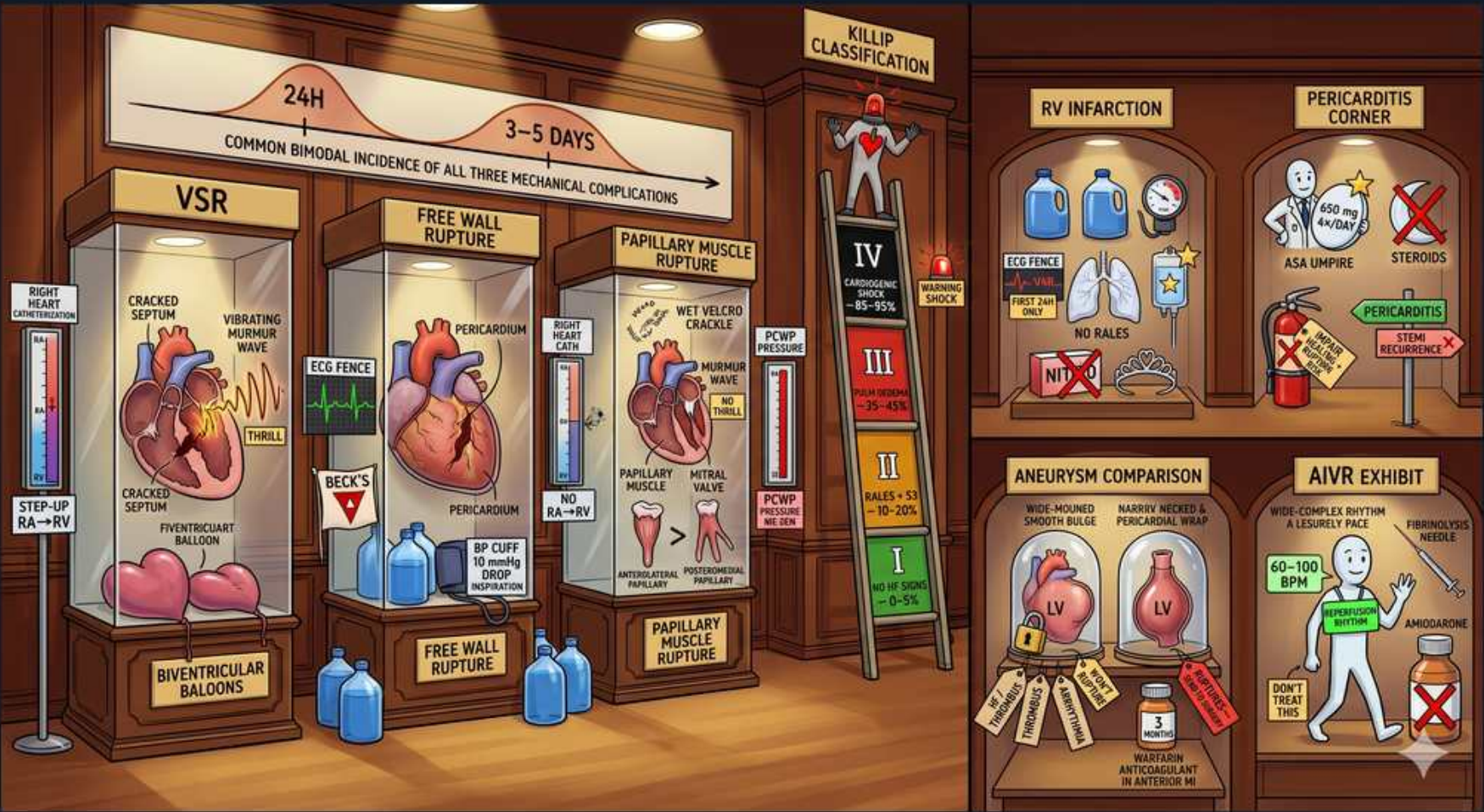


STEMI — Mechanical Complications & Killip Classification



1. Bimodal timing — 24h & 3-5 days

WHAT YOU SEE

Banner '24H ... 3-5 DAYS — common bimodal incidence of all three mechanical complications'.

WHAT IT ENCODES

Mechanical complications (VSR, free-wall rupture, papillary muscle rupture) peak bimodally: within 24h (early) and at 3-5 days (when necrotic tissue is weakest before scar forms). Reperfusion has reduced overall incidence.

BOARD TRAP

Day 3-5 is the classic 'rupture' window — sudden deterioration days after MI should trigger an echo for mechanical complication, not just re-MI.

2. VSR — ventricular septal rupture

WHAT YOU SEE

Cabinet 'VSR' with cracked septum and vibrating murmur waves.

WHAT IT ENCODES

Ventricular septal rupture creates a left-to-right shunt → harsh holosystolic murmur, often with a thrill, and an oxygen step-up from RA to RV. More common in anterior MI (LAD). Treat with surgical repair (± temporary mechanical support).

BOARD TRAP

VSR shows an O2 step-up RA→RV and a thrill; papillary muscle rupture does NOT step up and gives acute pulmonary edema with a softer murmur.

3. O2 step-up RA→RV / thrill

WHAT YOU SEE

'STEP-UP RA→RV' and 'THRILL' tags under VSR.

WHAT IT ENCODES

The hallmark of VSR is an oxygen saturation step-up between right atrium and right ventricle on right-heart catheterization, plus a palpable thrill. This distinguishes it from acute mitral regurgitation.

BOARD TRAP

O2 step-up = VSR. Acute MR from papillary rupture has NO step-up — use this to separate the two on the boards.

4. Free wall rupture — pericardium/Beck's

WHAT YOU SEE

Cabinet 'FREE WALL RUPTURE', pericardium, ECG fence, 'BECK'S'.

WHAT IT ENCODES

Free-wall rupture causes acute hemopericardium → tamponade with Beck's triad (hypotension, distended neck veins, muffled heart sounds), often pulseless electrical activity and sudden death. Emergent pericardiocentesis + surgery.

BOARD TRAP

Sudden PEA arrest or shock with tamponade physiology days after MI = free-wall rupture; it kills fast and needs the OR, not just pericardiocentesis.

5. BP cuff / 60 mmHg — tamponade hypotension

WHAT YOU SEE

BP cuff and 'NO RA→RV' tag under free-wall rupture.

WHAT IT ENCODES

Free-wall rupture produces tamponade with profound hypotension; unlike VSR there is no intracardiac shunt or O2 step-up. The presentation is obstructive shock/sudden collapse.

BOARD TRAP

No O2 step-up here — that finding belongs to VSR. Free-wall rupture = tamponade, not a shunt.

6. Papillary muscle rupture — acute MR

WHAT YOU SEE

Cabinet 'PAPILLARY MUSCLE RUPTURE', 'WET VELCRO CRACKLE', mitral valve.

WHAT IT ENCODES

Papillary muscle rupture causes acute severe mitral regurgitation → flash pulmonary edema (wet crackles), new soft systolic murmur, large V-waves and elevated PCWP. Posteromedial papillary muscle (single LAD or RCA supply) is most vulnerable.

BOARD TRAP

Posteromedial papillary muscle ruptures more often (single blood supply) — inferior MI is a classic setup, not anterior.

7. Giant V-wave / elevated PCWP

WHAT YOU SEE

'GIANT V-WAVE' and 'PCWP PRESSURE' / 'MITRAL VALVE' tags.

WHAT IT ENCODES

Acute MR generates tall regurgitant V-waves on the PCWP tracing and markedly elevated wedge pressure with flash pulmonary edema. Urgent surgical valve repair/replacement is required.

BOARD TRAP

Tall V-waves on PCWP point to acute MR (papillary rupture); they do not occur in VSR, which instead shows an O2 step-up.

8. Killip classification ladder

WHAT YOU SEE

Ladder 'KILLIP CLASSIFICATION' I-IV with a body at the top.

WHAT IT ENCODES

Killip class stratifies acute-MI mortality by exam: I = no failure, II = rales/S3, III = pulmonary edema, IV = cardiogenic shock. Higher class predicts steeply higher mortality.

BOARD TRAP

Killip is a clinical bedside score (rales, edema, shock) — don't confuse it with hemodynamic Forrester classes from PCWP/cardiac index.

9. Killip IV — cardiogenic shock 60-95%

WHAT YOU SEE

'IV CARDIOGENIC SHOCK 60-95%' rung.

WHAT IT ENCODES

Killip IV (cardiogenic shock) carries the highest mortality (historically 60-95%). Management: urgent revascularization, vasopressors/inotropes, and mechanical circulatory support.

BOARD TRAP

Cardiogenic shock is Killip IV with the worst prognosis — beta-blockers are contraindicated in this class.

10. RV infarction — clear lungs, no rales

WHAT YOU SEE

'RV INFARCTION' corner: jug, ECG fence, 'NO RALES'.

WHAT IT ENCODES

RV infarction (inferior STEMI) gives hypotension with CLEAR lungs and elevated JVP; treat with IV fluids and avoid nitrates/diuretics that drop preload. ECG: ST elevation in V4R.

BOARD TRAP

RV infarct = hypotension + clear lungs + raised JVP → give fluids; nitrates and diuretics are harmful here.

11. Pericarditis corner — ASA, not steroids

WHAT YOU SEE

'PERICARDITIS CORNER': ASA umpire, steroids crossed out.

WHAT IT ENCODES

Post-MI pericarditis (early peri-infarction or late Dressler) is treated with high-dose aspirin; NSAIDs and corticosteroids are avoided early because they impair myocardial healing and promote rupture/aneurysm.

BOARD TRAP

Use ASA for post-MI pericarditis — NSAIDs/steroids early can impair infarct healing and increase rupture/aneurysm risk.

12. True vs false aneurysm comparison

WHAT YOU SEE

'ANEURYSM COMPARISON': wide-mouth smooth (true) vs narrow-neck pericardial (false/pseudo).

WHAT IT ENCODES

True aneurysm = wide-mouthed, full-thickness, low rupture risk (causes HF, arrhythmia, mural thrombus). Pseudoaneurysm = contained rupture, narrow neck, HIGH rupture risk → needs surgery.

BOARD TRAP

Narrow-necked pseudoaneurysm is a contained rupture at high risk of bursting — operate; a wide-mouthed true aneurysm is far more stable.

13. AIVR exhibit — accelerated idioventricular rhythm

WHAT YOU SEE

'AIVR EXHIBIT': wide-complex rhythm, 'HR 60-100 BPM', 'REPERFUSION'.

WHAT IT ENCODES

Accelerated idioventricular rhythm (wide complexes, rate 60-100) is a benign reperfusion marker after successful thrombolysis/PCI. It is usually transient and needs no treatment.

BOARD TRAP

AIVR is benign and is NOT treated/suppressed — it signals successful reperfusion, not a dangerous VT.
